

COMP64501 Coursework: Network Design and Evaluation on the Fashion-MNIST Dataset

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I. INTRODUCTION

This report presents a compact residual network designed for the Fashion-MNIST dataset. The dataset is preprocessed using standardization and augmented with horizontal flipping to enhance generalization while preserving semantic structure. Hyperparameters were optimized using Bayesian optimization with k-fold cross-validation, yielding an average accuracy of approximately 93% with 67, 558 parameters. Ablation studies were performed on key architectural components to validate the design choices, and the model also exhibited strong generalization on the MNIST dataset with an accuracy of about 98%. Given that the network was trained on a single dataset and has a lightweight architecture, it may have limited adaptability to larger or more complex datasets in future tasks.

II. METHODS

A. Dataset and Preprocessing

The Fashion-MNIST dataset consists of ten balanced classes of $1 \times 28 \times 28$ grayscale images, containing 60,000 training samples and 10,000 test samples. The training set is further divided into training and validation subsets using an 8:2 split in subsequent stages of the workflow.

All images are standardized using the dataset statistics (mean $\mu = 0.286$ and standard deviation $\sigma = 0.353$), ensuring consistent pixel distributions and contributing to more stable optimization during training.

To enhance generalization while preserving the semantic integrity of the original images, horizontal flipping is applied as the primary data augmentation technique. The MNIST dataset is additionally employed as an external evaluation set to assess the generalization capability of the network and ensure that the architecture performs robustly on other datasets.

For a more rigorous assessment of the model's robustness, a 5-fold cross-validation is applied during the training process.

B. Network Architecture & Motivation

1) *Network Architecture:* As shown in Fig. 1, to better capture features from the low-dimensional input, a shallow architecture was adopted, consisting of a Stem layer, two sets of Residual Blocks [1], and downsampling transition layers.

After each convolution, Batch Normalization (BN) [2] is applied, followed by a ReLU activation [3]. Strided convolutions are employed for downsampling the feature maps.

2) *Channel Width:* Ablation experiments compared three channel configurations: 20-40-64, 16-32-64, and 32-64-128 (see Section III-B). The 20-40-64 configuration was ultimately selected, offering a good trade-off between model capacity and performance.

3) *Regularization:* Considering the limited number of training samples and the potential risk of overfitting, a combination of Global Average Pooling (GAP) and Dropout was employed. The Dropout rate before the classification head ($p = 0.12$) was determined via Bayesian optimization. In conjunction with the batch normal layers, this strategy effectively improves generalization on unseen data.

C. Hyperparameter Optimization

Bayesian optimization was employed to search for the best hyperparameters, implemented using the Hyperopt¹ library. The final configuration consisted of a batch size of 8, a learning rate of $lr = 0.0027$, a weight decay of 3.807×10^{-5} , and the SGD optimizer with momentum.

The number of training epochs was determined using a maximum-stop strategy (Fig. 2 a), the validation accuracy gradually plateaued around 80 epochs within a maximum of 200 epochs. Therefore, the model corresponding to the highest validation accuracy at 76 epochs was selected as the final configuration for subsequent experiments.

III. EXPERIMENT RESULTS

All experiments were conducted on a workstation running Ubuntu 20.04.6 with an Intel Core i9 CPU and an NVIDIA RTX 4060 GPU. CUDA 12.2 was used for GPU acceleration.

A. Training and Evaluation

The model was trained and evaluated using 5-fold cross-validation, yielding a mean accuracy of 93.11%, a standard deviation of 0.45%, and a coefficient of variation of 0.49%. The low variance across folds indicates stable performance and strong generalization, as illustrated in Fig. 3 (b).

The fold that achieved the highest validation performance (Fold 2) was further evaluated on the held-out test set, resulting

¹<https://github.com/hyperopt/hyperopt>

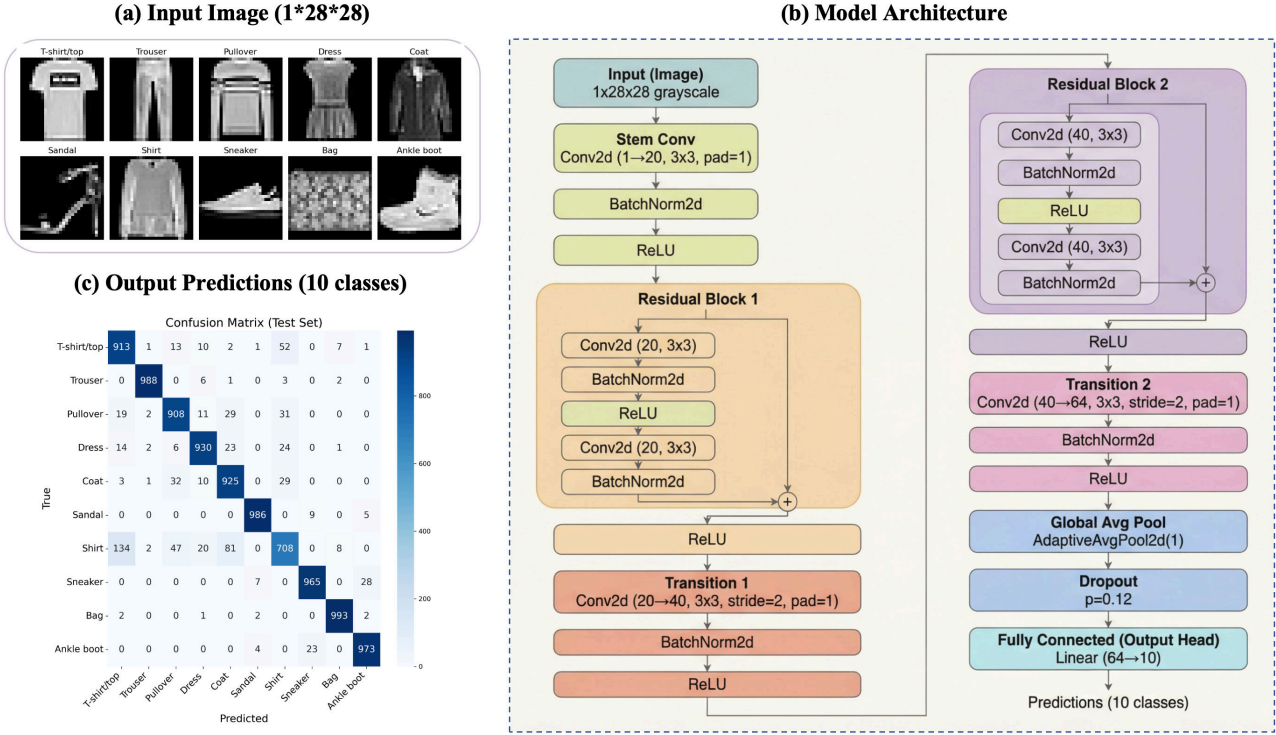


Fig. 1: Overview of the proposed classification framework. (a) Sample input images from the Fashion-MNIST dataset. (b) Detailed architecture of the custom Residual Network, featuring a stem convolution layer, two residual blocks, transition layers for downsampling, and a global average pooling (GAP) head. (c) Confusion matrix evaluated on the test set.

in a final accuracy of 92.89%. The corresponding confusion matrix is shown in Fig. 1 (b). It can be observed that the *shirt* category is more likely to be misclassified as *coat* or *pullover*, as shown in Fig. 3 (b).

To further assess the robustness of the proposed architecture, the model was also trained on the MNIST dataset. The model achieved a final accuracy of 98%, as shown in Fig. 3 (c-d). These results show that the proposed architecture generalizes effectively across datasets with different visual characteristics.

B. Ablation Study

To assess the impact of different design choices, a series of ablation experiments were conducted.

1) *Data Augmentation*: The experimental results indicate that the introduction of standardization and horizontal flipping is crucial for accelerating model convergence. As shown in Fig. 2 (b), models trained on unprocessed data plateau at a loss of around 0.2, whereas models with processed data exhibit a steady decline, ultimately reaching approximately 0.1.

2) *Channel Configuration*: The effect of channel width was evaluated using three configurations, as summarized in Table I. The 20-40-64 and 32-64-128 configurations achieved comparable accuracies (92.59% and 93.14%, respectively), while the 20-40-64 configuration required significantly fewer parameters (67,558 vs. 186,730). The 16-32-64 configuration also demonstrated strong performance, achieving 92.12% accuracy with

only 47,290 parameters. Considering the trade-off between model performance and parameter efficiency, the 20-40-64 configuration was selected for the final implementation. The validation accuracy progression across epochs is illustrated in Fig. 3 (a).

3) *Pooling Strategy*: In our experiments, Global Average Pooling consistently outperformed Global Max Pooling (92.77%). Unlike GMP, which focuses solely on peak activations, GAP aggregates information across the entire feature map, effectively capturing global structural patterns. This property is particularly important for distinguishing visual similarities and enhancing the model's generalization capability.

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REFERENCES

- [1] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," 2015. [Online]. Available: <https://arxiv.org/abs/1512.03385>
- [2] S. Ioffe and C. Szegedy, "Batch normalization: Accelerating deep network training by reducing internal covariate shift," 2015. [Online]. Available: <https://arxiv.org/abs/1502.03167>
- [3] V. Nair and G. E. Hinton, "Rectified linear units improve restricted boltzmann machines," in *Proceedings of the 27th International Conference on International Conference on Machine Learning*, ser. ICML'10. Madison, WI, USA: Omnipress, 2010, p. 807–814.

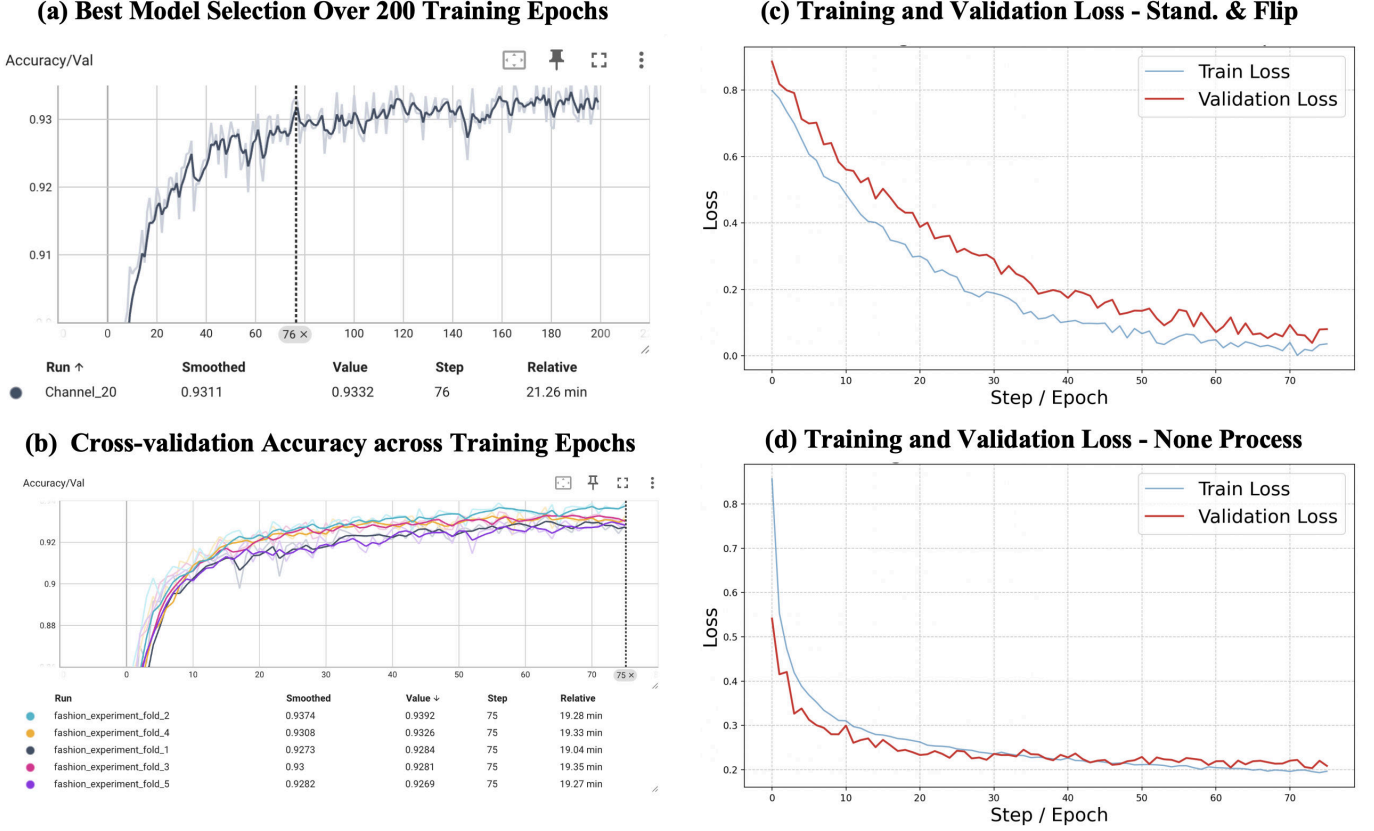


Fig. 2: Training metrics and ablation study. (a) Best model selection curve tracking validation accuracy. (b) Cross-validation accuracy across 5 folds indicates high stability. (c) vs. (d) Comparative loss curves showing the effectiveness of the "Standardization Flip" strategy

TABLE I: **Ablation study on data processing and model architecture.** The table reports Top-1 accuracy and parameter counts for each configuration. The performance difference (Δ) is computed relative to the baseline (Experiment 3).

ID	Data Augment	Channels	Pooling	Params	Top-1 Acc (%)	Δ
1	None	20-40-64	Avg	67,558	92.43	-0.68
2	Norm	20-40-64	Avg	67,558	92.08	-1.03
3	Norm + Flip (Baseline)	20-40-64	Avg	67,558	93.11	-
4	Norm + Flip	16-32-64	Avg	47,290	92.04	-1.07
5	Norm + Flip	32-64-128	Avg	186,730	93.14	+0.03
6	Norm + Flip	20-40-64	Max	67,558	92.15	-0.96

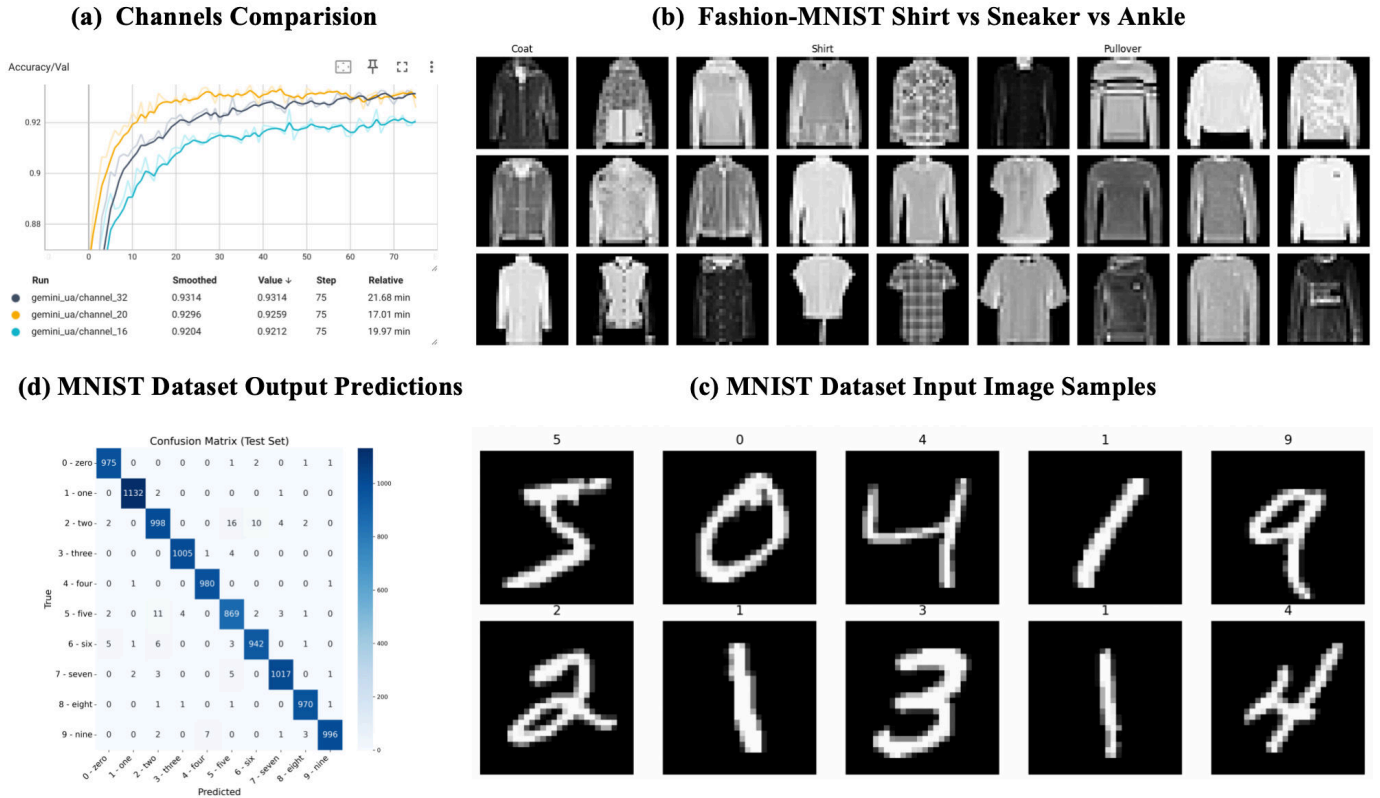


Fig. 3: Model performance on two datasets. The upper panel shows results on the Fashion-MNIST. Subfigure (a) compares validation accuracy for different base channel widths (16–32), highlighting the effect of network capacity on performance. Subfigure (b) illustrates challenging samples from frequently confused classes such as Coat, Shirt, and Pullover. The lower panel shows evaluation on the MNIST dataset as a supplementary dataset. Subfigure (c) presents representative input samples, and subfigure (d) shows the confusion matrix, demonstrating strong generalization with an accuracy of approximately 98%.